MATH 115AH: Week 1 (September 25 – 30, 2026)

Lecture notes for the first week of MATH 115AH (Linear Algebra, honors), Fall 2026: vector spaces and matrices, subspaces, fields and subfields, linear combinations and span, linear independence, bases, the Replacement Theorem and dimension, and a first look at linear maps.

1 September 25, 2026: Vector Spaces

1.1 The Example of $\mathbb{R}^2$

Example 1.1. $\mathbb{R}^2 = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : x \in \mathbb{R},\ y \in \mathbb{R} \right\}$, with

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} + \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ y_1 + y_2 \end{bmatrix} \tag{1}$$

and, for $a \in \mathbb{R}$,

$$a \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax \\ ay \end{bmatrix}. \tag{2}$$

[Sketch: the vector (x, y) drawn as an arrow from the origin in the plane, with its negative $(-x, -y)$.]

1.2 Definition of a Vector Space

Definition 1.2. A *vector space* (linear space) V over F ($\mathbb{R}$ or $\mathbb{C}$) is a set with 2 operations, addition and scalar multiplication, $(V, +, \cdot\,; 0)$, with a distinguished element 0, such that $\forall x, y \in V$, $\exists!\, x + y \in V$, and $\forall a \in F$, $x \in V$, $\exists!\, ax \in V$, satisfying

(i) $\forall x, y \in V$, $x + y = y + x$;

(ii) $\forall x, y, z \in V$, $(x + y) + z = x + (y + z)$;

(iii) $\forall x \in V$, $x + 0 = 0 + x = x$;

(iv) $\forall x \in V$, $\exists!\, y \in V$ s.t. $x + y = 0$;

(v) $\forall x \in V$, $1 \cdot x = x$ $(1 \in F)$;

(vi) $\forall a, b \in F$, $x \in V$, $a(bx) = (ab)x$;

(vii) $\forall a \in F$, $x, y \in V$, $a(x + y) = ax + ay$;

(viii) $\forall a, b \in F$, $x \in V$, $(a + b)x = ax + bx$.

Example 1.3. $\mathbb{R}$ is a vector space over $\mathbb{R}$. $\mathbb{C}$ is a vector space over $\mathbb{C}$.

Elements in V are *vectors*; elements in F are *scalars*.

Example 1.4. F^n, $n \geq 1$ integer:

$$F^n = \left\{ \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} : x_1, \ldots, x_n \in F \right\}. \tag{3}$$

For $\vec{x}, \vec{y} \in F^n$ and $\forall a \in F$,

$$\vec{x} + \vec{y} = \begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{pmatrix}, \qquad a\vec{x} = \begin{pmatrix} ax_1 \\ \vdots \\ ax_n \end{pmatrix}. \tag{4}$$

Remark. "0 and negation are unique."

1.3 Matrices

Example 1.5 (Matrices)**.** Let $m, n \geq 1$ be integers. An $m \times n$ *matrix* with entries $\in F$ is

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}. \tag{5}$$

An $n \times 1$ matrix is a column vector. For example,

$$\begin{bmatrix} 2+i & 2 \\ 3+2\sqrt{3} & 4 \end{bmatrix} \tag{6}$$

is 2×2 in $\mathbb{C}$. Then $M_{m\times n}(F) := \{m \times n \text{ matrices}/F\}$ is a vector space, with addition

$$A + B = \begin{bmatrix} a_{11} + b_{11} & \cdots & a_{1n} + b_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} + b_{m1} & \cdots & a_{mn} + b_{mn} \end{bmatrix} \tag{7}$$

and scalar multiplication

$$c \cdot A = \begin{bmatrix} ca_{11} & \cdots & ca_{1n} \\ \vdots & \ddots & \vdots \\ ca_{m1} & \cdots & ca_{mn} \end{bmatrix}. \tag{8}$$

How to relate to equations: from

$$\begin{cases} ax + by = c \\ dx + ey = f \end{cases} \tag{9}$$

we can get

$$\begin{bmatrix} a & b \\ d & e \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} c \\ f \end{bmatrix}. \tag{10}$$

1.4 Cancellation Law

Theorem 1.6 (Cancellation law for addition)**.** *Assume $x, y, z \in V$. If $x + z = y + z$, then $x = y$.*

Proof. For $z \in V$, there exists $v \in V$ s.t. $z + v = 0$. Then,

$$(x + z) + v = (y + z) + v \tag{11}$$
$$x + (z + v) = y + (z + v) \tag{12}$$
$$x = y. \tag{13}$$

□

1.5 Properties of Zero and Negatives

Theorem 1.7. *For the number $0 \in F$,*

1. *$0 \cdot v = 0$, $\forall v \in V$;*
2. *$(-a)x = -(ax) = a(-x)$, $\forall a \in F$, $\forall x \in V$;*
3. *$a \cdot 0 = 0$, $\forall a \in F$.*

Proof. (1) By the distributive law,

$$0 \cdot v + 0 \cdot v = (0 + 0)v = 0v \tag{14}$$

hence $0 \cdot v = 0$ (by cancellation).

(2) Using the distributive law, then (1),

$$(-a)x + ax = (-a + a)x = 0x = 0, \tag{15}$$

and using (3),

$$a(-x) + ax = a(-x + x) = a \cdot 0 = 0. \tag{16}$$

(3)

$$a \cdot 0 + a \cdot 0 = 0(a + a) = 0 \cdot 2a = 0a = 0. \tag{17}$$

□

1.6 Back to F

Back to F: $\in \mathbb{R}$ or $\mathbb{C}$; also $\mathbb{Q}$, and $\mathbb{Z}/p\mathbb{Z}$. For $\mathbb{Z}/p\mathbb{Z}$: $\forall a \in \mathbb{Z}$, $\exists$

$$a \cdot u + p \cdot v = 1 \tag{18}$$

for some $u, v \in \mathbb{Z}$ (since $(a, p) = 1$). Then, mod p,

$$\overline{a} \cdot \overline{u} = \overline{1}. \tag{19}$$

Also,

$$\underbrace{\overline{1} + \cdots + \overline{1}}_{p} = \overline{0}. \tag{20}$$

2 September 28, 2026: Subspaces

2.1 Recall

Recall $(V, +, \cdot)$ over $F = \mathbb{R}, \mathbb{C}$. V is a set, $0 \in V$,

$$+ : V \times V \to V \tag{21}$$
$$\cdot : F \times V \to V \tag{22}$$

s.t. $\forall x, y \in V$: $x + y = y + x$; $(x + y) + z = x + (y + z)$; $0 + x = x = x + 0$; $\exists!\, x' \in V$ s.t. $x + x' = 0$; and $\forall a, b \in F$, $x, y \in V$,

$$(a + b)(x + y) = a \cdot x + a \cdot y + b \cdot x + b \cdot y. \tag{23}$$

2.2 Subspaces and the Subspace Test

Definition 2.1. A subset W of V is called a *subspace* if W itself is a vector space over F with respect to $+|_W$, $\cdot|_W$.

Remark (Digest). Let $(V, +, \cdot\,; 0)$ and $W \subseteq V$. Then W is a subspace of V if

1. $\overline{0} \in W$;
2. W is closed under addition;
3. W is closed under scalar multiplication.

Example 2.2. $V = \mathbb{R}^2 = \left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} : x_1, x_2 \in \mathbb{R} \right\}$. Then $W = \left\{ \begin{bmatrix} x_1 \\ 0 \end{bmatrix} : x_1 \in \mathbb{R} \right\}$ is a subspace (✓), but $W = \left\{ \begin{bmatrix} x_1 \\ 1 \end{bmatrix} : x_1 \in \mathbb{R} \right\}$ is not (×).

Proof. (⇓) Since W is a vector space over F w.r.t. $+|_W$ and $\cdot|_W$, conditions 2 and 3 are given true, and $\exists\, \overline{0}_W \in W$ s.t.

$$\overline{0}_W + x = x, \quad \forall x \in W. \tag{24}$$

But

$$\overline{0} + x = x \quad \forall x \in W \subseteq V, \tag{25}$$

so

$$\overline{0} + x = x = \overline{0}_W + x \tag{26}$$
$$\overline{0} = \overline{0}_W. \tag{27}$$

(⇑) $\forall x, y \in W$, $a, b \in F$:

$$x + y = y + x, \quad (x + y) + z = x + (y + z) \tag{28}$$
$$x + \overline{0} = x \tag{29}$$
$$\exists\, (-x) \in V \text{ s.t. } x + (-x) = \overline{0}, \quad -x = -1 \cdot x \in W \text{ (by 3)} \tag{30}$$
$$1 \cdot x = x, \quad (a + b)(x + y) = ax + ay + bx + by \tag{31}$$
$$(ab)x = a(bx). \tag{32}$$

Therefore $\forall a, b \in F$, $x, y \in W \Rightarrow ax + by \in W$. □

Example 2.3.

$$\mathbb{R}[x] = \{\, a_0 + a_1 x + \cdots + a_n x^n : a_0, \ldots, a_n \in \mathbb{R} \,\} \tag{33}$$

$$\mathbb{R}[x]_{<m} = \{\text{polynomials of degree} < m\} \tag{34}$$

2.3 Transpose and Trace Examples

Example 2.4. $V = M_{m\times n}(F)$. Define the *transpose* of A to be

$$A^t = \begin{bmatrix} a_{11} & \cdots & a_{m1} \\ \vdots & \ddots & \vdots \\ a_{1n} & \cdots & a_{mn} \end{bmatrix}. \tag{35}$$

(i) $m = n$:

$$W_1 = \left\{\, A \in M_{m\times n}(F) : A = A^t \,\right\} \tag{36}$$

$$W_2 = \{\, A \in M_{m\times n}(F) : a_{ij} = 0,\ \forall i > j \,\} \tag{37}$$

$$W_3 = W_2^t \tag{38}$$

$$W_4 = \{\text{diagonal matrices}\} \tag{39}$$

$$W_4 = W_1 \cap W_2 \text{ and } W_1 \cap W_3 \text{ and } W_2 \cap W_3 \tag{40}$$

[Sketch: W_2 drawn as a square matrix with entries on and above the diagonal and 0 below it.]

For $m = n$: $\operatorname{tr}(A) = a_{11} + a_{22} + \cdots + a_{nn}$, and

$$W_5 = \{\, A \in M_{n\times n}(F) : \operatorname{tr}(A) = 0 \,\}. \tag{41}$$

2.4 Intersections of Subspaces

Theorem 2.5. *Any intersection of subspaces of V is a subspace of V.*

Proof. Let $\mathcal{C} = \{\text{subspaces of } V\}$ and

$$W_0 := \bigcap_{W\in\mathcal{C}} W. \tag{42}$$

$\overline{0} \in W$ for any $W \in \mathcal{C}$, so $\overline{0} \in \bigcap_{W\in\mathcal{C}} W = W_0$. $\forall a, b \in F$, $\forall x, y \in W_0 \subseteq W$, $ax + by \in W$ for any $W \in \mathcal{C}$, hence

$$ax + by \in \bigcap_{W\in\mathcal{C}} W = W_0. \tag{43}$$

□

Remark. The union of a subspace is only a subspace if one subspace is contained in the other.

3 September 29, 2026: Fields and Subfields

3.1 Basic Properties of Fields

Proposition 3.1. *Let F be a field. $\forall a \in F$, $0 \cdot a = 0$.*

Proof. First,

$$0 \cdot a + a = (0+1) \cdot a = 1 \cdot a = a \tag{44}$$

then,

$$0 = a + (-a) = (0 \cdot a + a) + (-a) \tag{45}$$
$$= 0 \cdot a + (a + (-a)) \tag{46}$$
$$= 0 \cdot a + 0 = 0 \cdot a. \tag{47}$$

□

Proposition 3.2. *Let F be a field. If $a, b \in F$ satisfy $a \cdot b = 0$, then $a = 0$ or $b = 0$.*

Proof. Assume $a \neq 0$. Then

$$b = 1 \cdot b = (a^{-1} \cdot a) \cdot b = a^{-1} \cdot (a \cdot b) \tag{48}$$
$$= a^{-1} \cdot 0 = 0. \tag{49}$$

□

3.2 Subfields

Definition 3.3. Let F be a field. A subset $E \subseteq F$ is called a *subfield* if

(i) $0, 1 \in E$;

(ii) $\forall a, b \in E$, $a + b \in E$, $a \cdot b \in E$;

(iii) $\forall a \in E$, $-a \in E$;

(iv) $\forall a \in E$, $a \neq 0$, $a^{-1} \in E$.

Let E be a subfield of F. By (ii) we get operations $+, \cdot : E \times E \to E$.

Exercise: show these operations make E into a field.

Example 3.4. $\mathbb{Q}(\sqrt{2}) = \left\{ a + b\sqrt{2} : a, b \in \mathbb{Q} \right\} \subseteq \mathbb{R}$ is a subfield of $\mathbb{R}$.

Proof. $0, 1 \in \mathbb{Q}(\sqrt{2})$. Clear: closure under $+$, closure under additive inverse. For $a, b, c, d \in \mathbb{Q}$,

$$(a + b\sqrt{2})(c + d\sqrt{2}) = ac + 2bd + (ad + bc)\sqrt{2} \in \mathbb{Q}(\sqrt{2}). \tag{50}$$

Say $a, b \in \mathbb{Q}$ s.t. $a + b\sqrt{2} \neq 0$; then a, b are not both 0, so $a - b\sqrt{2} \neq 0$. Then,

$$\frac{1}{a + b\sqrt{2}} = \frac{a - b\sqrt{2}}{(a - b\sqrt{2})(a + b\sqrt{2})} \tag{51}$$
$$= \frac{a}{a^2 - 2b^2} + \frac{(-b)}{a^2 - 2b^2}\sqrt{2} \in \mathbb{Q}(\sqrt{2}). \tag{52}$$

□

4 September 30, 2026 (Part 1): Linear Combinations and Span

4.1 Linear Combinations

Let $(V = \text{vector space}/F, +, \cdot; \overline{0})$. Given $v_1, \ldots, v_n \in V$ and $a_1, \ldots, a_n \in F$, then we get

$$a_1 v_1 + a_2 v_2 + \cdots + a_n v_n \in V. \tag{53}$$

Definition 4.1. Let S be a subset of V ($S \subseteq V$). A vector $v \in V$ is called a *linear combination of vectors in* S if $\exists\, u_1, \ldots, u_n \in S$, $a_1, \ldots, a_n \in F$ s.t.

$$v = a_1 u_1 + \cdots + a_n u_n. \tag{54}$$

Remark (Digest). If $S = \emptyset$, then no vector is a linear combination of vectors in S.

(i) Assume $S \neq \emptyset$; then $\overline{0} = 0 \cdot u$ for any $u \in S$, so $\overline{0}$ is a linear combination.

(ii) If v, v' are linear combinations of vectors $\in S$ and $c, c' \in F$, what about $cv + c'v'$?

For (ii), say

$$v = a_1 u_1 + \cdots + a_n u_n \tag{55}$$
$$v' = a_1' u_1' + \cdots + a_m' u_m' \tag{56}$$

then consider $\{u_1, \ldots, u_n, u_1', \ldots, u_m'\} = \{u_1, \ldots, u_N\}$, $N \leq n + m$. Then

$$cv + c'v' = (ca_1 + c'a_1')u_1 + \cdots + (ca_N + c'a_N')u_N \tag{57}$$

is a linear combination of vectors in S.

4.2 Span Is a Subspace

Theorem 4.2 (Theorem/Definition)**.** *Define* $\text{Span}(S)$ *as the set of all linear combinations of vectors in* S*. Assume* $S \neq \emptyset$*. Then* $\text{Span}(S)$ *is a subspace of* V*.*

5 September 30, 2026 (Part 2): Span and Linear Independence

5.1 Examples of Span

Example 5.1. In $\mathbb{R}^3$, $u_1 = (1, 0, 1)$, $u_2 = (0, 1, 1)$:

$$3u_1 - 2u_2 = (3, -2, 1) \in \text{Span}(\{u_1, u_2\}). \tag{58}$$

Example 5.2. In $\mathbb{R}^2$: [Sketch: two non-parallel vectors u_1, u_2 drawn from the origin.] $\text{Span}(\{u_1, u_2\}) = \mathbb{R}^2$.

Example 5.3. In F^n, let

$$u_1 = (x_1^{(1)}, \ldots, x_n^{(1)}), \quad u_2 = (x_1^{(2)}, \ldots, x_n^{(2)}), \quad u_3 = (x_1^{(3)}, \ldots, x_n^{(3)}). \tag{59}$$

Given $v = (y_1, \ldots, y_n) \in F^n$, how to check if $v \in \text{Span}(\{u_1, u_2, u_3\})$? Equivalently, $\exists\, a, b, c \in F$ s.t. $v = au_1 + bu_2 + cu_3$. Then, write

$$v = \begin{bmatrix} u_1 & u_2 & u_3 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = au_1 + bu_2 + cu_3 \tag{60}$$

$$\begin{bmatrix} x_1^{(1)} & x_1^{(2)} & x_1^{(3)} \\ \vdots & \vdots & \vdots \\ x_n^{(1)} & x_n^{(2)} & x_n^{(3)} \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}. \tag{61}$$

Definition 5.4. We say that S *generates* V if $V = \text{span}(S)$.

Example 5.5. In $\mathbb{R}^2$, $\{u_1, u_2\}$ generates $\mathbb{R}^2$. [Sketch: u_1 along the horizontal axis and u_2 along the vertical axis.]

Example 5.6. In F^n, let

$$e_1 = (1, 0, \ldots, 0) \tag{62}$$
$$e_2 = (0, 1, \ldots, 0) \tag{63}$$
$$e_n = (0, 0, \ldots, 1). \tag{64}$$

Then $F^n = \text{span}(\{e_1, e_2, \ldots, e_n\})$, since any $v = (x_1, \ldots, x_n) \in V$ is $x_1 e_1 + \cdots + x_n e_n$.

5.2 Linear Independence

Definition 5.7 ("redundancy")**.** A subset $S \subseteq V$ is called *linearly independent* if $\forall u_1, \ldots, u_n \in S$ and $a_1, \ldots, a_n \in F$, if we have

$$a_1 u_1 + \cdots + a_n u_n = 0 \tag{65}$$

then $a_1 = \cdots = a_n = 0$.

S is *linearly dependent* if $\exists\, u_1', \ldots, u_n' \in S$ and $a_1', \ldots, a_n' \in F$ not all 0 s.t. $a_1' u_1' + \cdots + a_n' u_n' = 0$.

Example 5.8. In F^n, $S = \{e_1, e_2, \ldots, e_n\}$. If $a_1, \ldots, a_n \in F$ s.t. $a_1 e_1 + \cdots + a_n e_n = 0$, then $(a_1, \ldots, a_n) = (0, \ldots, 0)$, so $a_1 = \cdots = a_n = 0$: l.i.

Example 5.9. In $\mathbb{R}^3$, $S = \{e_1, e_2, e_1 + 2e_3\}$. If $a_1, a_2, a_3 \in \mathbb{R}$ s.t.

$$a_1 e_1 + a_2 e_2 + a_3(e_1 + 2e_3) = 0 \tag{66}$$
$$(a_1, 0, 0) + (0, a_2, 0) + (a_3, 0, 2a_3) = 0 \tag{67}$$
$$(a_1 + a_3, a_2, 2a_3) = (0, 0, 0) \tag{68}$$

hence $a_1 = a_2 = a_3 = 0$: l.i. Or,

$$(a_1 + a_3)e_1 + a_2 e_2 + 2a_3 e_3 = 0 \tag{69}$$
$$a_1 + a_3 = a_2 = 2a_3 = 0 \tag{70}$$

hence $a_1 = a_2 = a_3 = 0$.

HW: V; $S = \{v_1, v_2, v_3\}$ l.i. Prove: $v_1, v_2, v_1 + 2v_3$ l.i.

6 September 30, 2026 (Part 3): Bases

6.1 Proof of the HW Statement

Proof. If $a_1v_1 + a_2v_2 + a_3(v_1 + 2v_3) = 0$, then

$$(a_1 + a_3)v_1 + a_2v_2 + 2a_3v_3 = 0 \tag{71}$$

and by l.i., $a_1 + a_3 = a_2 = 2a_3 = 0$, hence $a_1 = a_2 = a_3 = 0$. □

6.2 Removing a Dependent Vector

Example 6.1. In $\mathbb{R}^4$,

$$u_1 = (1, 3, -4, 2), \quad u_2 = (2, 2, -4, 0), \quad u_3 = (1, -3, 2, -4). \tag{72}$$

Then

$$4u_1 - 3u_2 + 2u_3 = 0 \tag{73}$$
$$u_3 = -\tfrac{1}{2}(4u_1 - 3u_2) \tag{74}$$

hence $\text{Span}(\{u_1, u_2, u_3\}) = \text{span}(\{u_1, u_2\})$.

6.3 Adding a Vector to a Linearly Independent Set

Theorem 6.2. *Let $\emptyset \neq S \subseteq V$ be l.i., and $v \in V$ s.t. $v \notin S$. Then $S \cup \{v\}$ is l.i. $\iff$ $v \notin \text{span}(S)$.*

Proof. ($\Rightarrow$) Assume that $v \in \text{Span}(S)$. Then $v = a_1u_1 + \cdots + a_nu_n$ for $a_1, \ldots, a_n \in F$ and $u_1, \ldots, u_n \in S$. Then

$$a_1u_1 + \cdots + a_nu_n + (-1) \cdot v = 0 \tag{75}$$

so $S \cup \{v\}$ is l.d., a contradiction.

($\Leftarrow$) Assume

$$a_1u_1 + \cdots + a_nu_n + a_{n+1}v = 0 \tag{76}$$

for $a_1, \ldots, a_{n+1} \in F$ and $u_1, \ldots, u_n \in S$. If $a_{n+1} = 0$, then $a_1u_1 + \cdots + a_nu_n = 0$, and since S is l.i., $a_1 = \cdots = a_n = 0$. If $a_{n+1} \neq 0$, then

$$v = -\frac{1}{a_{n+1}}(a_1u_1 + \cdots + a_nu_n) \in \text{Span}(S), \tag{77}$$

a contradiction. □

Recall: $(V, +, \cdot)/F$, $S \subseteq V$,

$$\text{span}(S) = \{ \textstyle\sum a_iu_i : u_i \in S \}, \qquad \text{span}(\emptyset) = \{0\}. \tag{78}$$

$\{v_1, \ldots, v_n\} \subseteq V$ is l.i. iff $\forall a_1, \ldots, a_n \in F$, $\sum_{i=1}^n a_iv_i = 0 \Rightarrow a_1 = \cdots = a_n = 0$.

Theorem 6.3. *Let $S \subseteq V$ be l.i., $v \in V$. Then $S \cup \{v\}$ is l.i. $\iff$ $v \notin \text{span}(S)$.*

6.4 Bases

Definition 6.4. A subset $\beta \subseteq V$ is called a *basis* for V if

(i) β is l.i.;

(ii) V is the span(β).

Remark (Digest). In $\mathbb{R}^{\text{[illegible]}}$: $\{e_1, e_2\}$ ✓, $\{e_1, e_3\}$ ✓, $\{e_1, e_2, e_3\}$ ×.

Example 6.5. F^n: $\{e_1, \ldots, e_n\}$, the *standard basis*.

Example 6.6. $M_{m\times n}(F)$: $\{ E_{ij} : 1 \le i \le m,\ 1 \le j \le n \}$, where E_{ij} is the $m \times n$ matrix s.t. the (i, j) entry is 1 and the rest are 0.

Example 6.7. $\{\overline{0}\}$ is NOT l.i., since $1 \cdot \overline{0} = \overline{0}$. $\emptyset$ is a basis of $\{\overline{0}\}$.

6.5 Unique Representation

Theorem 6.8. *For distinct vectors $u_1, \ldots, u_n \in V$, the set $\beta = \{u_1, \ldots, u_n\}$ is a basis of $V \iff \forall v \in V$ there is a unique expression of v as a combination of $u_1, \ldots, u_n$, i.e. $\forall v \in V$, $\exists!\, a_1, \ldots, a_n \in F$ s.t.*

$$v = \sum_{i=1}^{n} a_i u_i. \tag{79}$$

Proof. ($\Downarrow$) $V = \text{span}(\beta)$, so $v = \sum_{i=1}^n a_i u_i$ for some $a_i \in F$. For uniqueness, assume

$$v = \sum a_i u_i = \sum a_i' u_i \tag{80}$$

then

$$\sum (a_i - a_i') u_i = \overline{0} \tag{81}$$

and by l.i., $a_i - a_i' = 0$, i.e. $a_i = a_i'$.

($\Uparrow$) $V = \text{span}(\beta)$ ✓, and $\overline{0} = 0 \cdot u_1 + \cdots + 0 \cdot u_n$. Assume $a_1 u_1 + \cdots + a_n u_n = \overline{0}$. Uniqueness of the expression of $\overline{0}$ implies that $a_1 = 0, a_2 = 0, \ldots, a_n = 0$. So β is l.i. □

6.6 A Basis from a Finite Generating Set

Theorem 6.9. *If S is a finite set that generates V, a subset of S is a basis for V.*

Proof. Let $S = \{u_1, \ldots, u_n, \ldots, u_N\}$, $V = \text{span}(S)$.

(i) S_1: is $\{u_1\}$ l.i., i.e. is $u_1 \neq 0$? Keep if l.i.

(ii) S_2: is $\{u_1, u_2\}$ l.i.? Keep u_2 if yes.

(iii) S_3: is $\{u_1, u_2, u_3\}$ l.i.? Keep u_3 if yes.

Continuing, we get $S_N \subseteq S$, with $\text{span}(S_1) = \text{span}(\{u_1\})$, $\text{span}(S_2) = \text{span}(\{u_1, u_2\})$, …, s.t.

$$\text{span}(S_N) = \text{span}(\{u_1, \ldots, u_N\}) = \text{span}(S) = V, \tag{82}$$

and S_N is l.i. by our construction. So S_N is a basis for V. □

6.7 The Replacement Theorem and Dimension

Theorem 6.10 (Replacement Theorem $(*)$)**.** *Let $S \subseteq V$ s.t. $V = \text{Span}(S)$, $|S| = n$. Let $L \subseteq V$ be l.i., $m = |L|$. Then $|L| \leq n$, and $\exists\ n-m$ vectors H s.t.* $\text{span}(L \cup H) = V$.

Remark (Upshot). If V has a basis consisting of n vectors, then any basis for V has n vectors. So $n := \dim(V)$ is intrinsic.

Proof of upshot assuming the theorem. Let β be a basis for V with n vectors, and let β' be a basis for V. Applying $(*)$ to $S = \beta$ and $L = \beta'$,

$$|\beta'| \leq |\beta| = n. \tag{83}$$

Applying $(*)$ to $S = \beta'$ and $L = \beta$,

$$|\beta| \leq |\beta'| \tag{84}$$

hence $|\beta'| = |\beta|$. □

Proof of $(*)$. Let $L = \{v_1, \ldots, v_m\}$.

Base step: $m = 0$, we can take $H = S$.

Induction: assume the theorem is proved for $|L'| = 1, \ldots, m-1$ (so $m - 1 \leq n$). Now for L, consider $L' = \{v_1, \ldots, v_{m-1}\}$. The induction hypothesis gives $m - 1 \leq n$ and $\exists\, H_{m-1} = \{w_1, \ldots, w_{n-(m-1)}\}$ s.t. $V = \text{span}(L' \cup H_{m-1})$. So

$$v_m = \sum_{i=1}^{m-1} a_i v_i + \sum_{j=1}^{n-(m-1)} b_j w_j. \tag{$\dagger$}$$

If $n-(m-1) = 0$, then $v_m \in \text{span}(v_1, \ldots, v_{m-1})$, a contradiction to L being l.i. So $n-(m-1) \geq 1$, hence $m \leq n$.

At least one $b_j \neq 0$ (by $(\dagger)$). Say $b_1 \neq 0$ (up to renumbering); then $(\dagger)$ implies

$$w_1 = \frac{1}{b_1} v_m - \sum_{i=1}^{m-1} \frac{a_i}{b_1} v_i - \sum_{j=2}^{n-(m-1)} \frac{b_j}{b_1} w_j. \tag{85}$$

Say we can take

$$H = \{w_2, \ldots, w_{n-(m-1)}\} \tag{86}$$

with $|H| = n - m$ and $V = \text{span}(L \cup H)$. □

Theorem 6.11. *Let $W \subseteq V$ be a subspace. Then* $\dim(W) \leq \dim(V)$*, and* $\dim W = \dim V$ *if and only if $W = V$.*

6.8 Linear Maps

Definition 6.12 (Maps)**.** Let V, W be vector spaces over F. $T : V \to W$ is a *linear transformation* if

$$T(c_1 x + c_2 y) = c_1 T(x) + c_2 T(y) \quad \forall c_1, c_2 \in F,\ x, y \in V. \tag{87}$$

Equivalently, $T(x + y) = T(x) + T(y)$ and $T(cx) = cT(x)$.

Example 6.13. 1. $T(\overline{0}) = \overline{0}$.

2. $T : \mathbb{R}^3 \to \mathbb{R}^2$, $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \mapsto \begin{pmatrix} 2x_1 - x_3 \\ 4x_2 + x_3 \end{pmatrix}$.

4. $T : \mathbb{R}^2 \to \mathbb{R}^2$, rotation by θ.

5. $T : \mathbb{R}^n \to \mathbb{R}^m$, projection to the first m elements.

6. $T : M_{m\times n}(F) \to M_{n\times m}(F)$, $A \mapsto A^t$:

$$T(c_1A + c_2B) = (c_1A + c_2B)^t = (c_1A)^t + (c_2B)^t \tag{88}$$
$$= T(c_1A) + T(c_2B). \tag{89}$$

7. $T : \mathbb{R} \to \mathbb{R}^2$, $x \mapsto \begin{bmatrix} x \\ 2x \end{bmatrix}$.

Example 6.14 $(*)$**.** For $A \in M_{m\times n}(\mathbb{R})$, $T_A : \mathbb{R}^n \to \mathbb{R}^m$, $x \mapsto Ax$.